



大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY



ENN 新奥

基于变分矩方法的高保真快速平衡求解器

VEQ : Veloce/Variational EQUilibrium Solver

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背景



MHD 平衡是聚变等离子体物理分析的基础（输运、稳定性等）

传统代码的局限性

- ✗ 面向“单次求解”
- ✗ 依赖固定的离散表示

新场景需求

- ✓ 需作为反复、高频调用的基础算子
- ✓ 支撑大规模参数扫描、实验重建、ML 训练
- ✓ 更快的平衡位形求解

变分矩平衡 (VEQ)

数学：参数化、光滑、可微 → 便于插值、求导、重采样及 AI 耦合

计算：高保真 & 高效率并存 → 快速且大规模的数据生成



大纲



1. 参数化平衡和变分形式
2. `veqpy`: 固定边界平衡的变分求解器
3. 基准测试
4. 实际应用



参数化平衡



$$\begin{aligned} R(\rho, \theta) &= R_0 + a [h(\rho) + \rho \cos \bar{\theta}], \\ Z(\rho, \theta) &= Z_0 + a [v(\rho) - \rho \kappa(\rho) \sin \theta]. \end{aligned}$$

极向: Fourier

$$\begin{aligned} \bar{\theta}(\rho, \theta) &= \theta + c_0(\rho) + \sum_{m=1}^M c_m(\rho) \cos(m\theta) \\ &\quad + \sum_{n=1}^N s_n(\rho) \sin(n\theta), \end{aligned}$$

$$\begin{aligned} c_m &= \rho^m \left[c_{ma} + (1 - \rho^2) \sum_{l=0}^L c_{ml} T_l(x) \right], \\ s_n &= \rho^n \left[s_{na} + (1 - \rho^2) \sum_{l=0}^L s_{nl} T_l(x) \right], \end{aligned}$$

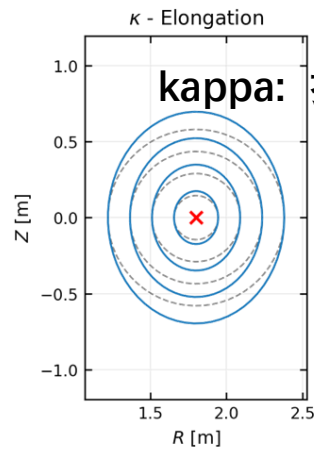
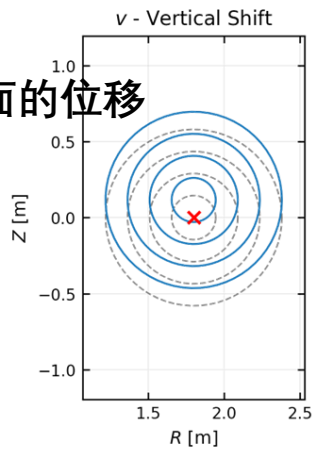
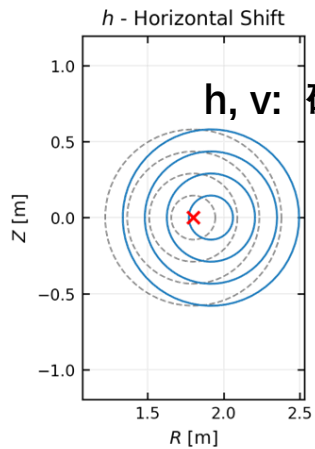
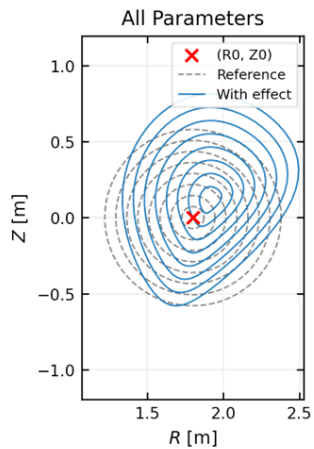
径向: Chebyshev

$$\begin{aligned} x &= 2\rho^2 - 1, \\ T_0(x) &= 1, \quad T_1(x) = x, \\ T_{l+1}(x) &= 2xT_l(x) - T_{l-1}(x). \end{aligned}$$

$$\begin{aligned} h &= (1 - \rho^2) \sum_{l=0}^L h_l T_l(x), \\ v &= (1 - \rho^2) \sum_{l=0}^L v_l T_l(x), \\ \kappa &= \kappa_a + (1 - \rho^2) \sum_{l=0}^L \kappa_l T_l(x), \end{aligned}$$

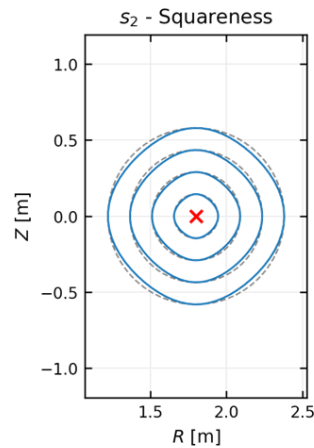
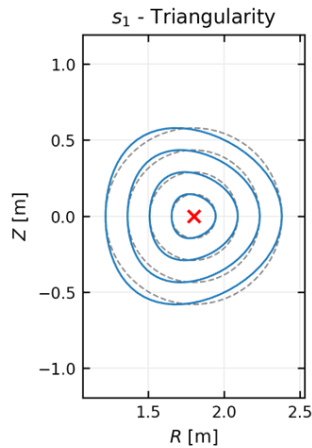
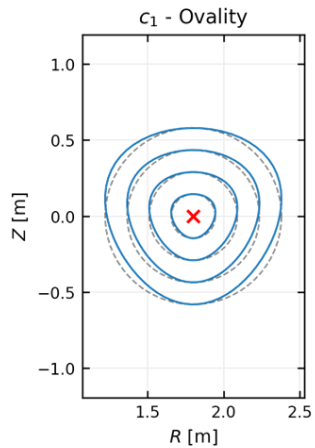
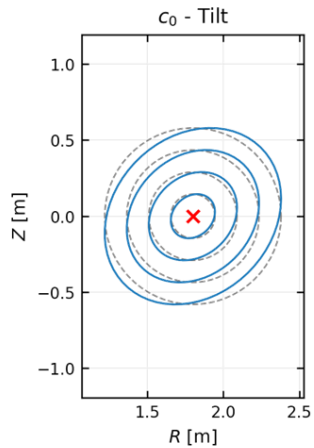


参数化平衡



h, v : 磁面的位移

κ : 拉长比



c, s : Fourier 修正



参数化平衡的变分形式



Lagrangian density^[1]:
$$L = \frac{|\nabla\psi|^2}{2\mu_0 R^2} - \left(\frac{F^2}{2\mu_0 R^2} + P \right)$$

定义 Grad-Shafranov 算子:
$$\hat{G} = \frac{J}{R} (FF_\psi + \mu_0 R^2 P_\psi) + \left(\frac{g_{\theta\theta}\psi_\rho}{JR} \right)_\rho - \left(\frac{g_{\rho\theta}\psi_\rho}{JR} \right)_\theta$$

对流变分:
$$\delta_{\mathbf{r}}\psi + \nabla\psi \cdot \delta\mathbf{r} = \left(\frac{\partial\psi}{\partial p_k} + \nabla\psi \cdot \frac{\partial\mathbf{r}}{\partial p_k} \right) \delta p_k = 0$$

强形式:
$$\hat{G} = 0,$$

等价于GS方程

弱形式:

$$\int_0^1 d\rho \int_0^{2\pi} d\theta \left[\hat{G} \cdot \left(\frac{\partial\hat{\psi}}{\partial R} \frac{\partial R}{\partial p_k} + \frac{\partial\hat{\psi}}{\partial Z} \frac{\partial Z}{\partial p_k} \right) \right] = 0$$

robust
$$\int_0^1 d\rho \int_0^{2\pi} d\theta \left(\hat{G} \cdot \frac{\partial\hat{\psi}}{\partial\psi_k} \right) = 0$$

robust
$$\int_0^1 d\rho \int_0^{2\pi} d\theta \left(\hat{G} \cdot \frac{\partial\hat{F}}{\partial F_k} \right) = 0$$

P_k (待优化的参数)



残差向量

robust: 难以分离出 psi 或 F 显式表达式的情况



多种输入模式



Route	输入剖面
PF (Strict)	P_ψ, FF_ψ
PP (Strict)	P_ψ, ψ_ρ
PI (Strict)	P_ψ, I_{tor}
PJ1 (Strict)	P_ψ, j_{tor}
PJ2 (Robust)	$P_\psi, j_{\parallel}$
PQ (Robust)	P_ψ, q

Pprime, FFprime

Pprime, psi

Pprime, I_{tor}

$$I_{\text{tor}} = \int_0^\rho j_{\text{tor}} S_\rho d\rho,$$

Pprime, j_{tor} (面平均 j_{phi})

Pprime, j_{para}

$$j_{\parallel} = \frac{\langle \mathbf{j} \cdot \mathbf{B} \rangle}{\langle \mathbf{B} \cdot \nabla \phi \rangle}$$

Pprime, q

支持 rho 或 psi 坐标下的输入



veqpy: 固定边界平衡的变分法求解

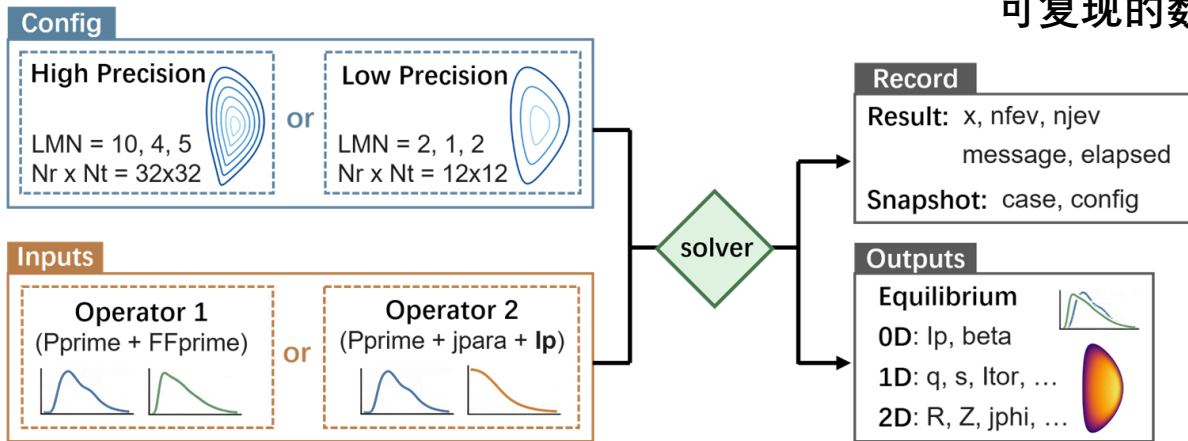


veqpy: A Python package for tokamak MHD equilibrium calculations, serving the **VEQ (Veloce/Variational EQUilibrium)**^[2] project.

code: github.com/zhangtakeda/veqpy-dev

参数量 + 求解网格 + 优化算法

可复现的数据组合

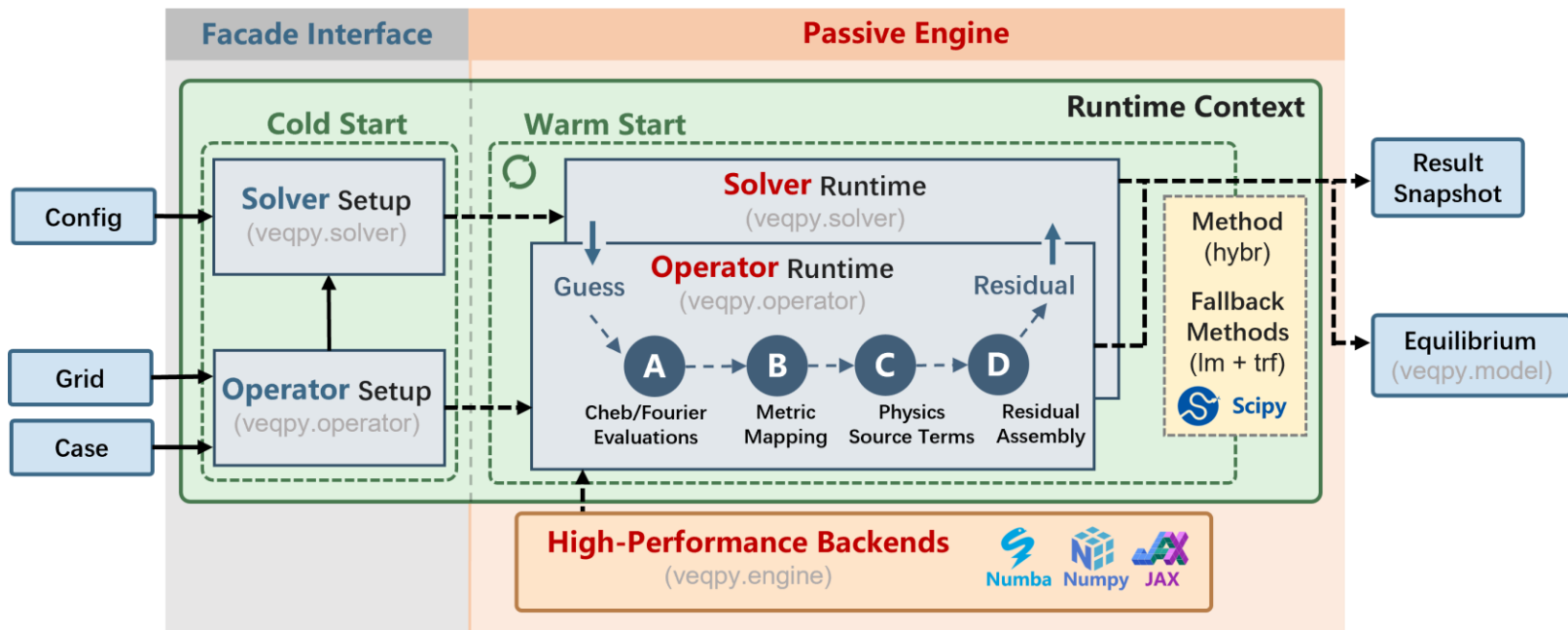


边界 + 输入剖面 + 电流/比压约束

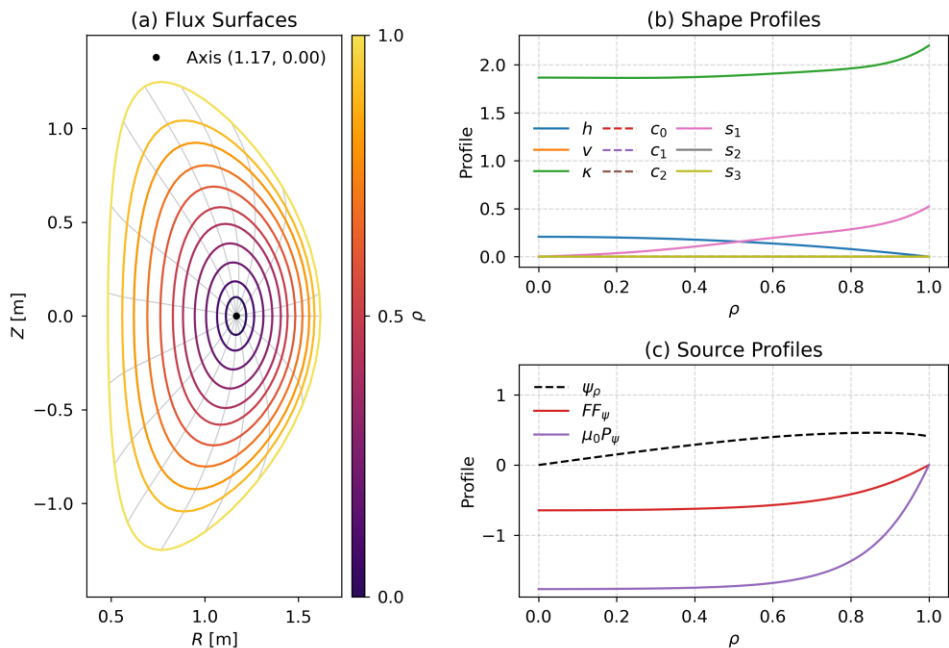
参数化, 光滑的平衡类型

1. 输入检查 and 无量纲化
2. 内存分配和预计算

1. 在已有内存块上执行计算
2. 在 (求解配置, 输入模式) 组合中执行具体逻辑



Demo: 用于程序内部的物理一致性检验



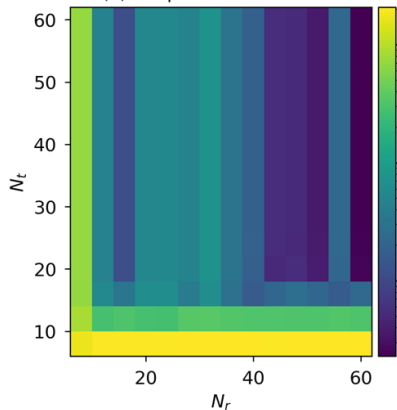
$$FF_\psi = (1 - \beta_0)\alpha_f \frac{\exp(\alpha_f \hat{\psi}) - \exp(\alpha_f)}{1 + (\alpha_f - 1) \exp(\alpha_f)},$$

$$\mu_0 P_\psi = \beta_0 \alpha_p \frac{\exp(\alpha_p \hat{\psi}) - \exp(\alpha_p)}{1 + (\alpha_p - 1) \exp(\alpha_p)}.$$

R_0, Z_0, a, B_0	1.05 m, 0 m, 0.568 m, 3.0 T
$R_0/a, \kappa, \bar{\delta}$	1.85, 2.2, 0.50
I_p	3.0 MA
$\beta_0, \alpha_p, \alpha_f$	0.75, 5.0, 3.32
(N_r, N_θ)	64 × 64, Legendre
$(L_{\hat{\psi}}, L_h, L_\kappa, L_{s1})$	(5, 3, 5, 3)

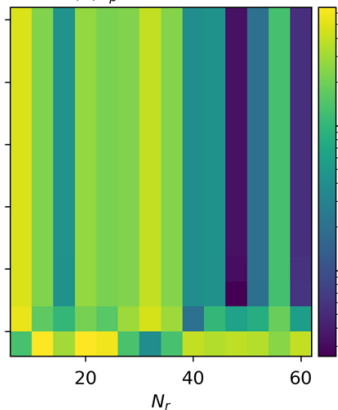
形状参数 Err

(a) Shape Parameter Error



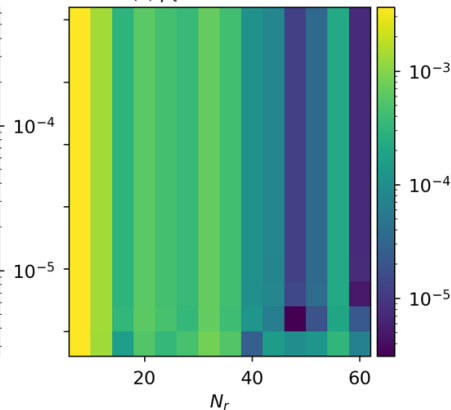
总电流 Err

(b) I_p Relative Error



比压 Err

(c) β_t Relative Error

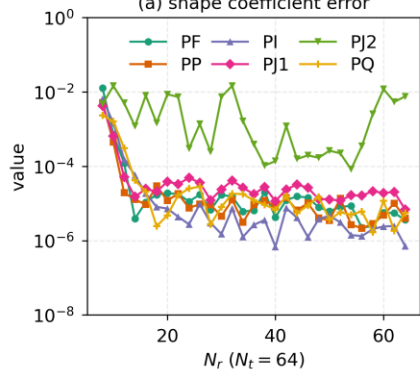


PF-route:

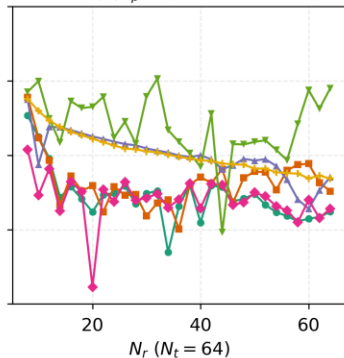
不同 (N_r , N_t) 网格下的收敛行为

N_t 达到阈值后对精度基本无影响

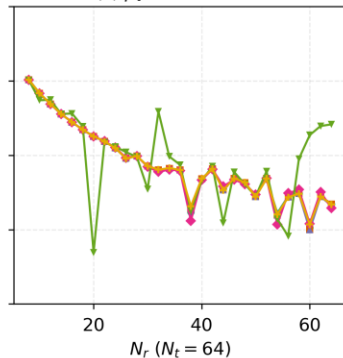
(a) shape coefficient error



(b) I_p relative error



(c) β_t relative error



各种 route:

不同 (N_r) 网格下的收敛行为

各模式求解基本一致,
差分/积分导致 PJ2 存在数值误差

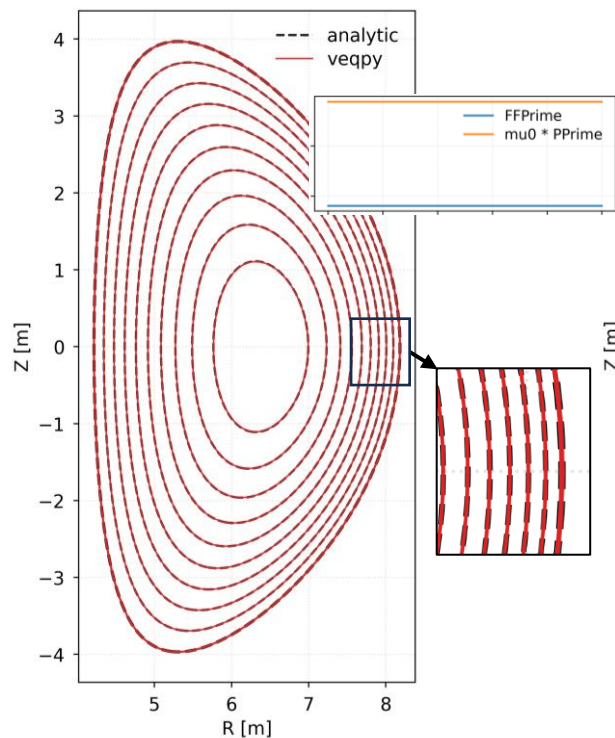


实际应用：多种等离子体位形求解



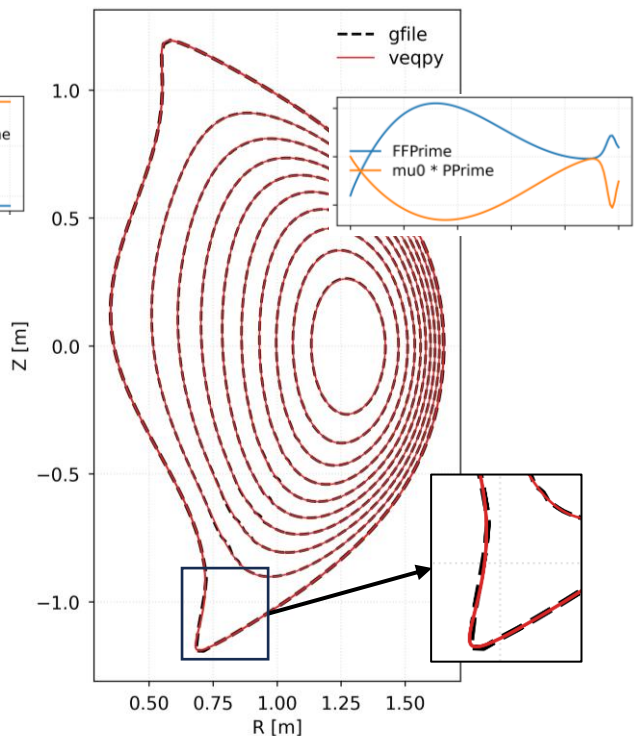
ITER case: D-shape

Solov'ev 解析解



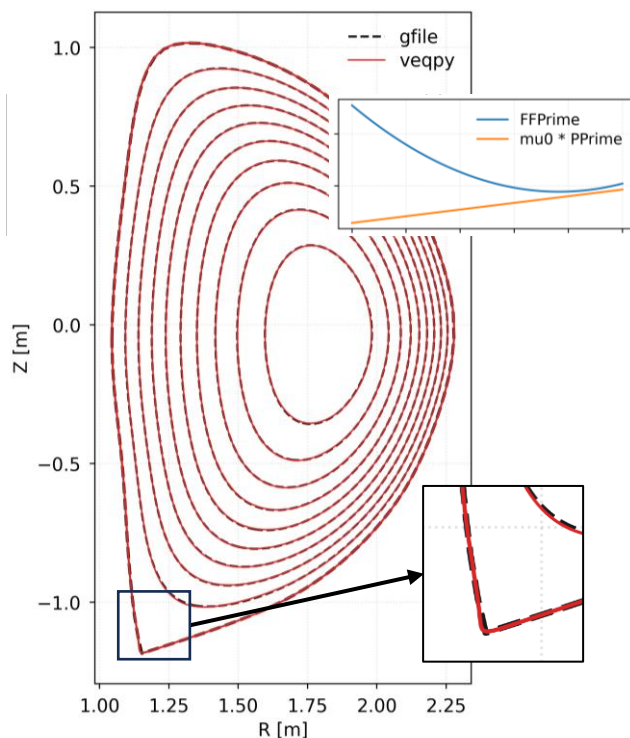
ST case: H-mode

CHEASE(有限元)



DIII-D case: X-point

EFIT(差分/Green)





实际应用：参数量对误差的影响

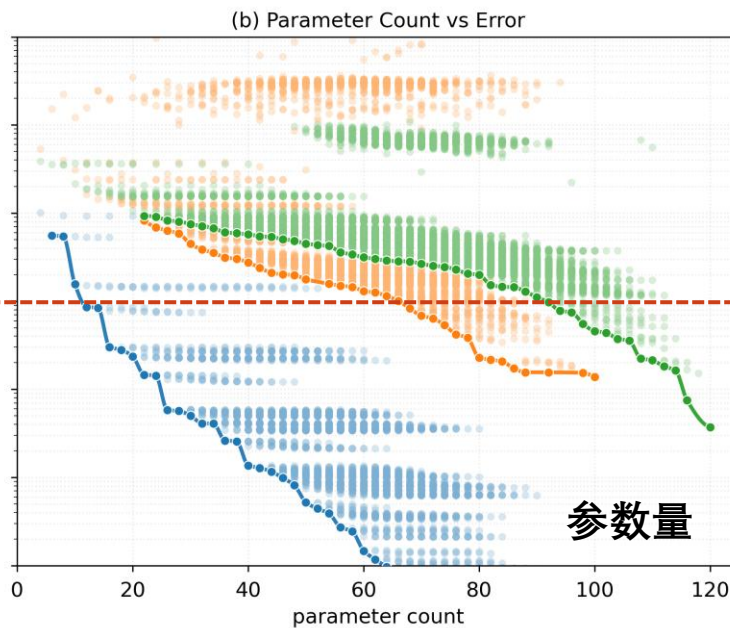
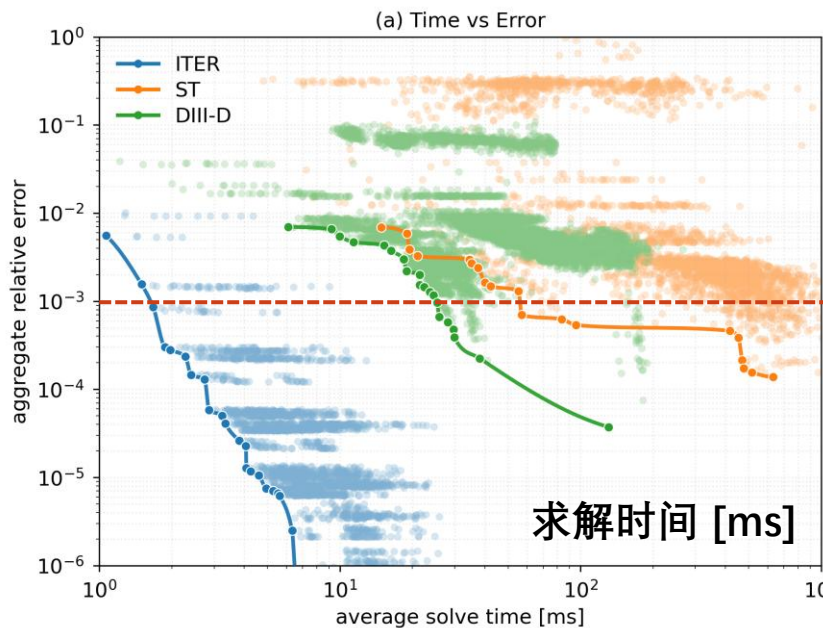


ITER (D-shape) : 12 参数 (1.67 ms, 0.86 mm) → 谱精度收敛

ST (H-mode) : 74 参数 (57.0 ms, 0.70 mm)

DIII-D (X-point) : 94 参数 (25.3 ms, 0.97 mm)

使用 32x32 的网格



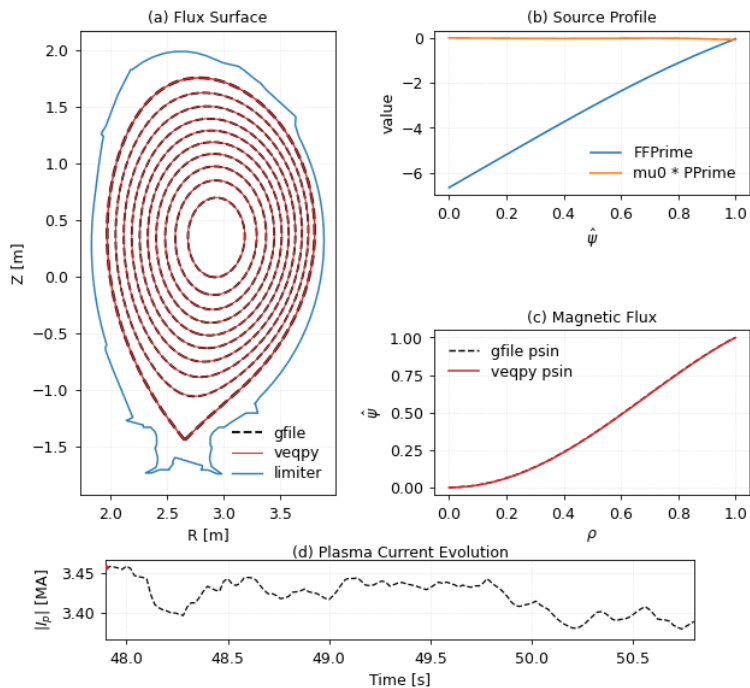


总结与讨论



veqpy@JET

使用 EFIT++ 反演的结果并进行对比



1. **veqpy** 具备多种输入模式且在物理上自治
2. 复杂形状, X-point也能和传统 FEM/Green 程序匹配
3. 能够在 1~100 ms 给出 误差 < 1 mm 的解 (CPU+单线程)

自由边界/平衡反演

边界参数加入优化, 额外的实验数据作为最小二乘约束

加速计算

提供高质量初始值, 或作为算子加入 Picard 迭代

快速生成高保真数据 → 代理模型

光滑可微的平衡表述, 能快速生成大量在任意网格下的数据

局限性

bean-shaped/多连通的等离子体? 局部不平衡?



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Thanks for your attention!

veqpy : github.com/zhangtakeda/veqpy-dev

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Related Work :

- 利用快速谱求解器 VEQ-R 研究环向旋转对球形环平衡的影响 (李星宇)
- 含与不含平衡求解器的托卡马克系统程序针对新奥球形环的结果比较 (陈蔡雪)
- 三维平衡超快变分参数法求解器 VEQ-3D 进展 (孟炜棋)
- ASTRA 中三矩平衡求解器 EMEQ 准确性分析 (赵睿)